

1 DPD ENGINE *Mirheo*

2 SURROGATES + SENSITIVITY *PyTorch + SALib*

3 BAYESIAN CALIBRATION *Korali*

4 DATA-INFORMED EMB MODELS

Definity

$\mathcal{M}_{d_1}$
 $\mathcal{M}_{d_2}$
 $\mathcal{M}_{d_3}$

Generate training data

S_{d_1}
 S_{d_2}
 S_{d_3}

Sensitivity

NL = 0

$S_{d_1}^0$
 $S_{d_2}^0$
 $S_{d_3}^0$

Independent inference

θ_{d_1}
 θ_{d_2}
 θ_{d_3}

Hierarchical inference

$\tilde{\theta}_{d_1}$
 $\tilde{\theta}_{d_2}$
 $\tilde{\theta}_{d_3}$

$\mathcal{M}_{d_1}(\tilde{\theta}_{d_1})$
 $\mathcal{M}_{d_2}(\tilde{\theta}_{d_2})$
 $\mathcal{M}_{d_3}(\tilde{\theta}_{d_3})$

Validation

Compression
+ acoustic
experiments

$\mathcal{A}_{d_1-d_3}$

SonoVue

$\mathcal{M}_{d_4}$
 $\mathcal{M}_{d_5}$
 $\mathcal{M}_{d_6}$

Generate training data

S_{d_4}
 S_{d_5}
 S_{d_6}

Sensitivity

NL = 0

$S_{d_4}^0$
 $S_{d_5}^0$
 $S_{d_6}^0$

Independent inference

θ_{d_4}
 θ_{d_5}
 θ_{d_6}

Hierarchical inference

$\tilde{\theta}_{d_4}$
 $\tilde{\theta}_{d_5}$
 $\tilde{\theta}_{d_6}$

$\mathcal{M}_{d_4}(\tilde{\theta}_{d_4})$
 $\mathcal{M}_{d_5}(\tilde{\theta}_{d_5})$
 $\mathcal{M}_{d_6}(\tilde{\theta}_{d_6})$

Validation

Indentation
+ acoustic
experiments

$\mathcal{A}_{d_4-d_6}$